
🔦 SplitLight: An
Exploratory Toolkit for
Recommender Systems
Datasets and Splits

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Recommender Systems, Information
Retrieval, LLM, Fundamental models

Author/co-author: **20+ papers** on top
conferences and journals

Current:
Sber, HSE University, MSU

Education:
MSU, YSDA

Sber AI Lab:

Developing technologies & solutions for the Bank

Advancing Science

120+ publication A*/A from 2019

Driving Business Value

Gen AI transformation

Technologies

Multi-agent systems

Self-critical AI

Collaborative AI

Attack-resilient AI

Proactive LLMs

Graphs in LLMs



Sber
AI Lab

Where Research Meets
Business Impact



Business Transformation

AI-Solutions for Bank

Bots & contact centers

GenAI in RecSys

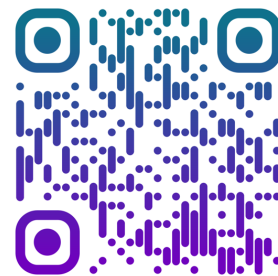
Healthcare

Education

DS Agent

Text2Agent

10+ Open-Source Products Released



GitHub sb-ai-lab



RePlay

A recommendation framework model building and evaluation



Py-Boost

Python-based gradient boosting library



Tsururu

A time series forecasting strategies framework



Sim4Rec

Simulator for training and evaluation of Recommender Systems



LightAutoML

Fast and customizable framework for automatic ML model creation



RIDE

Accelerate Dijkstra's algorithm on diverse graph structures



PyTorch-LifeStream

PyTorch for building embeddings on discrete event sequences using self-supervision



ESGify

Multilabel news classification with respect to ESG risks



SIRIN

LLM hallucination detection

Open Source datasets for RecSys

Zvuk

The dataset of listening to music tracks from the Zvuk service for a period of 4 months

Interactions: 244,673,551

Users: 382,790

Items: 1,506,950



MegaMarket

The dataset on purchases on the MegaMarket marketplace for a period of 4 months

Interactions: 196,644,020

Users: 2,730,776

Items: 3,562,321



SBOL

The dataset contains records on offers and purchases of banking products over the period of 4 month

interactions: 1,056,889

users: 190,439

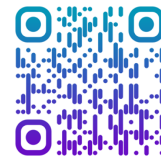
items: 19



<https://www.kaggle.com/datasets/alexxtl>

Multimodal Banking Dataset

Contains more than 1.5M corporate clients with several modalities



Huggingface



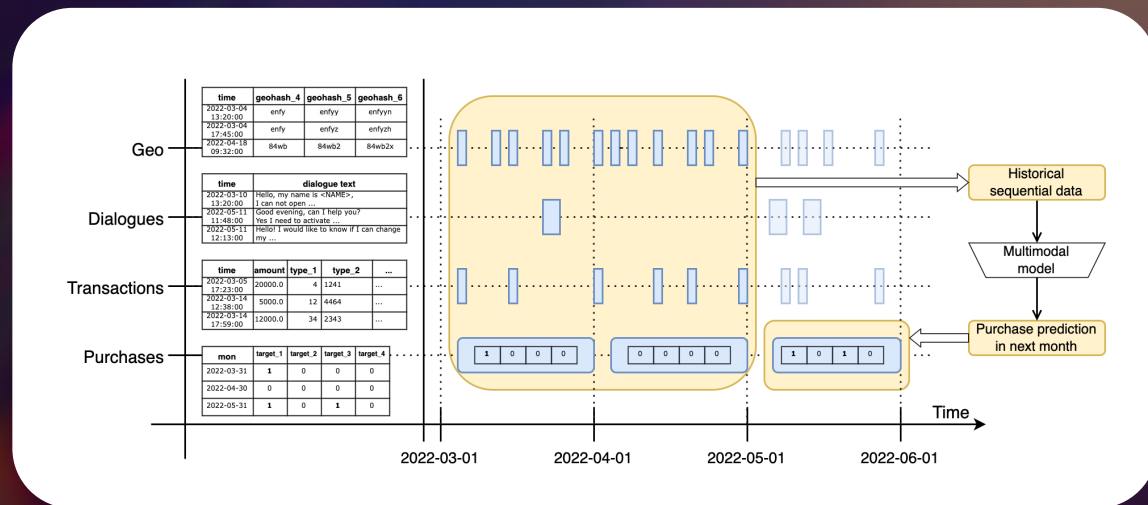
MBD-mini

Task:

- purchase prediction in the next month

Modalities:

- Bank transactions
- Embeddings of dialogues
- Geo position events



Open Source in scientific papers

Scientific paper



Open source code



Open source tool



Scientific paper



What are Recommender Systems?

RecSys features

- ✓ A large number of users
- ✓ A large number of items
- ✓ Significantly less interactions than $\#users * \#items$
- ✓ Users are not able to view all items

							
	5		4		5		
	4	5		5		3	
		4				5	4
	5	5					

Sequential Recommendations



Ivy



Timeline



Lily



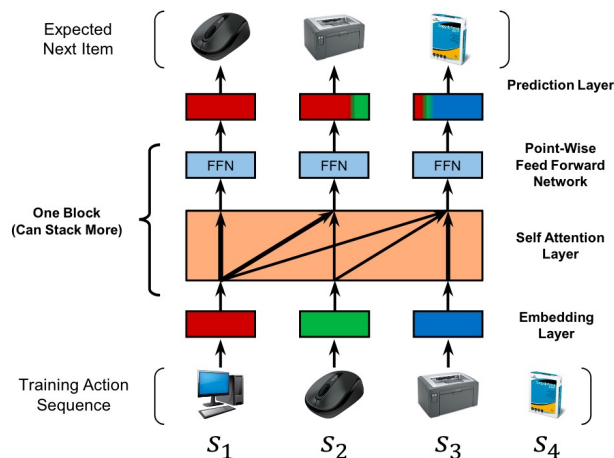
?

Timeline

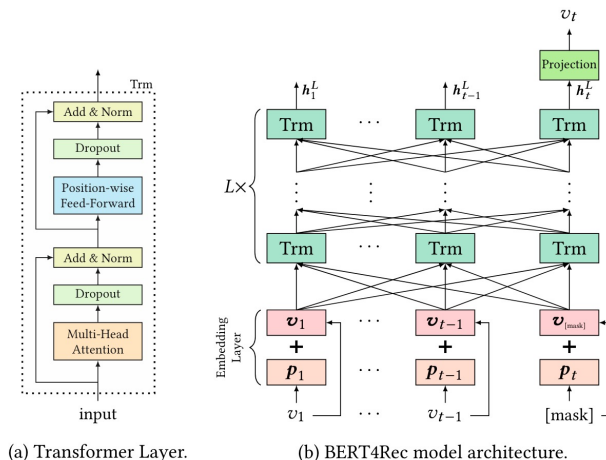
Recommendations



The most popular sequential algorithms

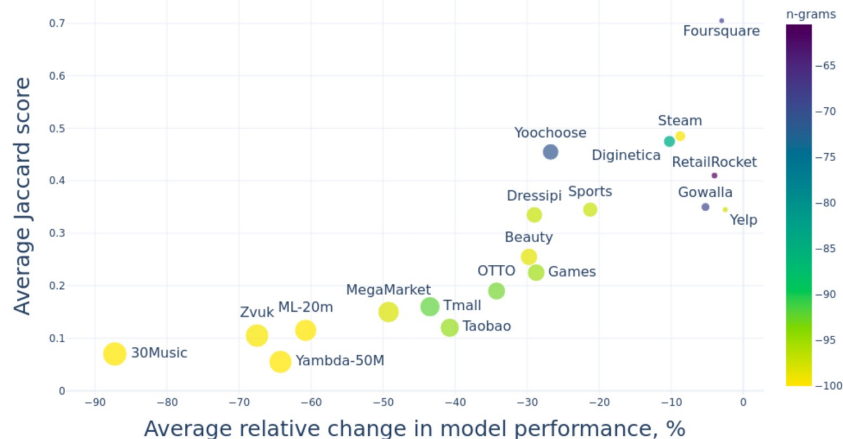


Kang, Wang-Cheng, and Julian McAuley.
"Self-attentive sequential recommendation."
2018 IEEE international conference on data mining
(ICDM). IEEE, 2018.



Sun, Fei, et al. "BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer." Proceedings of the 28th ACM international conference on information and knowledge management. 2019.

Problems in experimental protocols

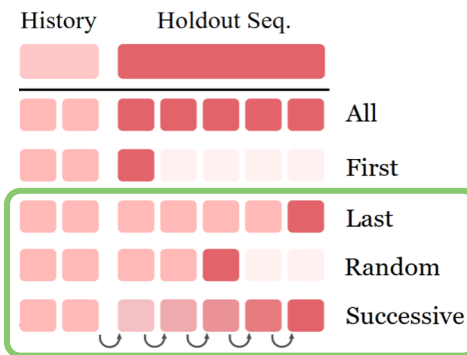


An Analysis of Sequential Patterns in Datasets for Evaluation of Sequential Recommendations

<https://doi.org/10.1145/3787969>

Time to Split: Exploring Data Splitting Strategies for Offline Evaluation of Sequential Recommenders

<https://doi.org/10.1145/3705328.3748164>





SplitLight: Motivation

- 🙋 Why different papers using the same dataset and data split report completely different metrics?
- 🙋 Why a method works in a paper, but I am not able to reproduce given the model and hyperparams?
- 🙋 Why A/B and offline results are low-correlated?



Possible reason:

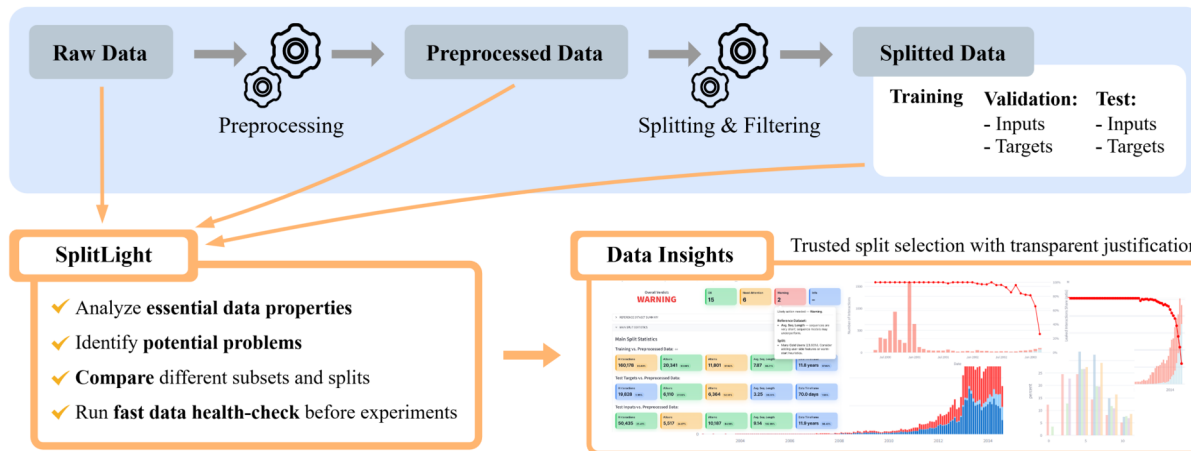
Data quality and data pre-processing and splitting pipelines details

- 🕵️ Are there quality issues in my dataset and split?
- 🕵️ What is specific about my data and should be handled in data preprocessing?
- 🕵️ Is externally preprocessed and split data reliable and free from major errors?
- 🕵️ How similar is my dataset to the latest real data properties?



SplitLight: Goals

Data Splitting Pipeline



Analyse datasets and splitting results to:

- understand what data you have on hands
- avoid common pitfalls in preprocessing and splitting
- choose proper filters and the other preprocessing steps
- compare different splitting results to choose the best split
- collect, check and report all necessary statistics to enhance transparency and reproducibility of your research



SplitLight: Who Is the User?

Academic and industrial practitioners, who want to...



Understand the data to design preprocessing and splitting pipelines



Analyse, how different preprocessing and splitting steps affect the data, and compare

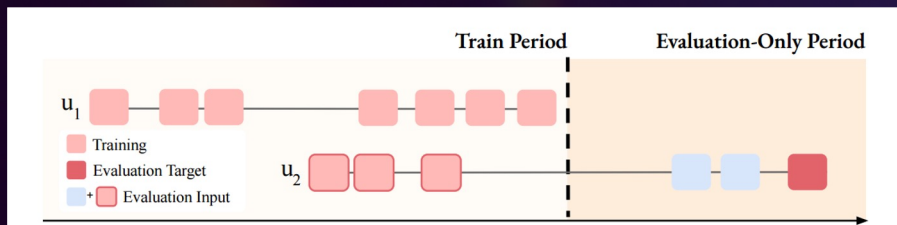


Document the data statistics to enhance reproducibility

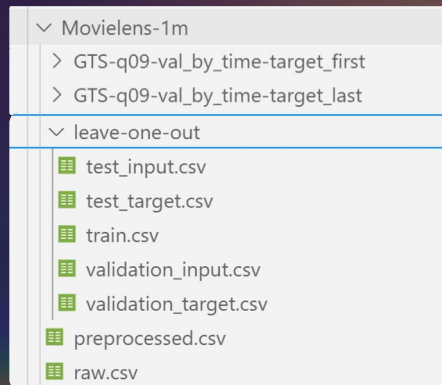


Audit the results of external (black-box) preprocessing and splitting

SplitLight: Data Layout



Evaluation input and target definition using the example of Global Temporal Split with last-item target. Evaluation inputs include all within-user history before evaluation targets.



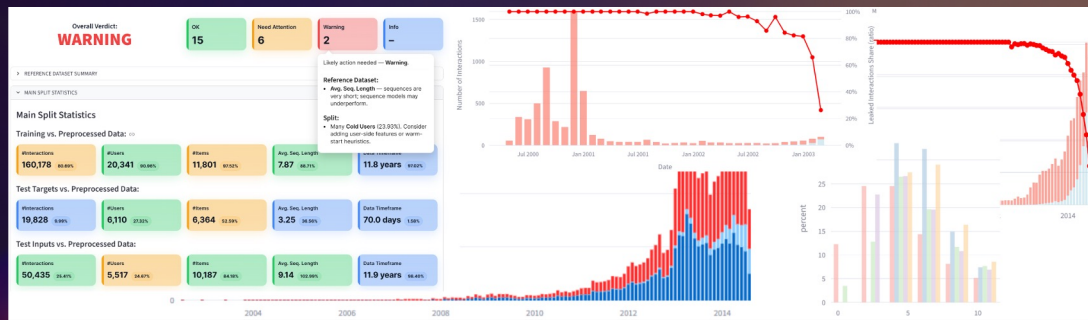
SplitLight data folder structure

-  **Check your data:** user id, item id, timestamp columns are required. Add dataset config with column names' mapping for raw data analysis.
-  **Preprocess and split** the data using your own setup or use illustrative pipelines from SplitLight
-  **Store** raw, preprocessed and split data in parquet/csv into data folder
-  **Analyse!**



SplitLight: Usage

Run easy-to-use **Streamlit** UI for analysis in a no-code mode



Indeed, we managed to eliminate data leakage!

But what about data statistics?

```
[121... compare_subsets([test_target, test_input, train], preprocessed, subset_names=['test target', 'test input', 'train'])
```

	n_users		n_items		n_interactions		avg_seq_length		density		timeframe	
	Abs. value	%	Abs. value	%	Abs. value	%	Abs. value	%	Abs. value	%	Abs. value	%
test target	6346.0	10.36	13885.0	54.25	48591.0	10.00	7.656949	96.56	0.000551	177.99	7.778496e+05	5.92
test input	216.0	0.35	1182.0	4.62	1541.0	0.32	7.134259	89.97	0.006036	1948.12	9.427770e+02	0.01
train	54648.0	89.18	25575.0	99.93	433360.0	89.19	7.930025	100.01	0.000310	100.08	1.235612e+07	94.08

Go into details in Jupyter Notebooks



SplitLight: Functionality

SplitLight

Summary

Dataset and Split

Core and Temporal Statistics

Interactions Over Time

Repeat Consumption

Split Analysis

Temporal Leakage

Cold Start

Compare Splits

Core and Temporal Statistics

Time Deltas

Item Positions

Dataset and Subsets

- Analyze and compare raw and preprocessed data in terms of core and temporal statistics
- Identify repeated consumption patterns
- Visualize interactions distribution over time

Subsets and Splits

- Analyze splitted data in terms of core and temporal statistics and compare subsets with full data
- Identify and visualize presence of data leakage
- Quantify and visualize user and item cold start

Compare splits

- Compare different splits in terms of core and temporal statistics
- Identify distributions shifts for target subset

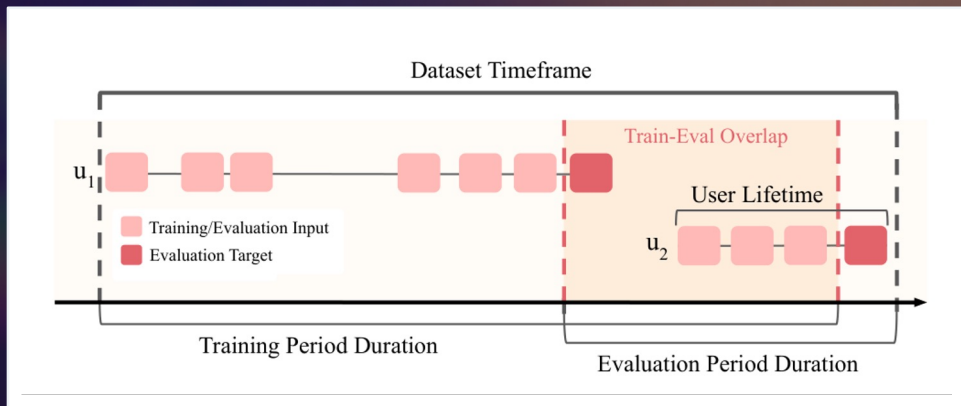
Core and temporal statistics

CORE

Interactions, users, items · item popularity ·
sequence length · density

TEMPORAL — THE KEY ADDED DIMENSION

- Dataset timeframe: first → last interaction span
- Time delta Δt : gap between consecutive user events
- Timestamp collisions: same timestamp within a user — makes splitting non-deterministic
- User & item lifetime: how fast the user base / catalog turns over

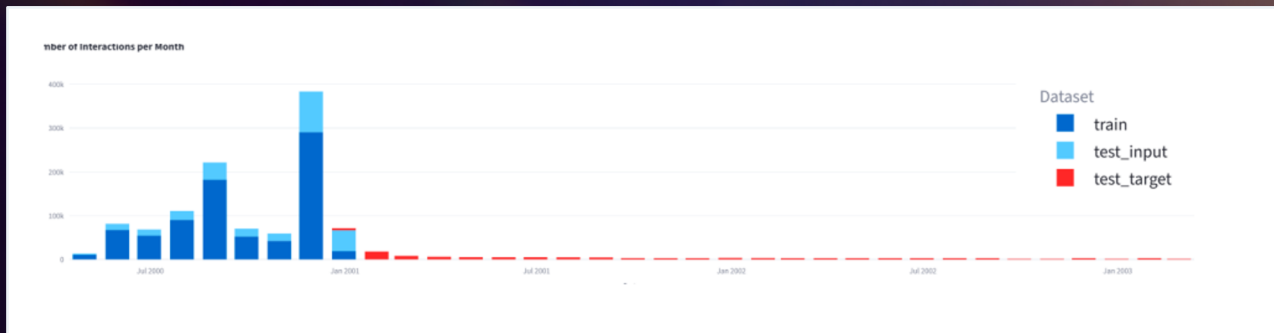


Temporal properties under a leave-one-out split: timeframe, train/eval durations, overlap, and per-user lifetime.



Why time matters: orders sequences correctly, exposes data collection issues, and shows how close an open dataset is to your in-house data

Interactions over time: spotting skew & overlap



Temporal skew in ML-1M: under a standard GTS $q_{0.9}$ split, the test target subset spans nearly 76% of the dataset timeframe.

Data-collection gaps

missing or interrupted logging
affect evaluation

Activity skew

skew across time distorts the
split closeness-to-reality

Subset overlap

training and evaluation periods
overlap each other

Repeat consumption patterns

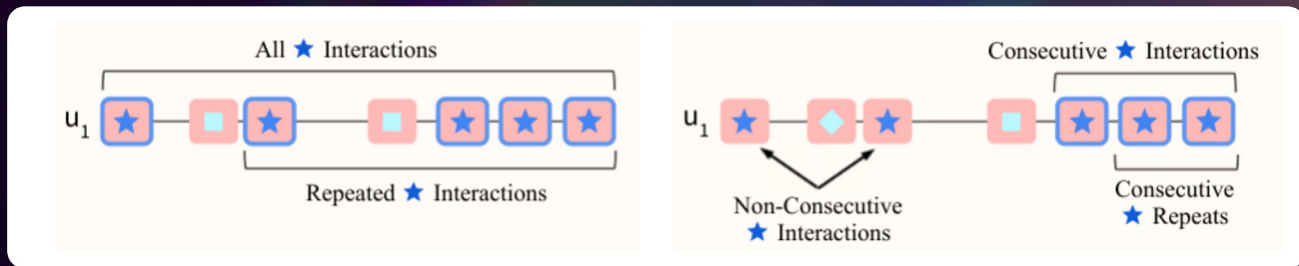
Repeats are rare in one-time consumption, but common in streaming or grocery

Repeated interactions

Any occurrence of an item after its first appearance in a user's history.

Consecutive repeats

The same item appearing in immediate succession — analyzed separately.

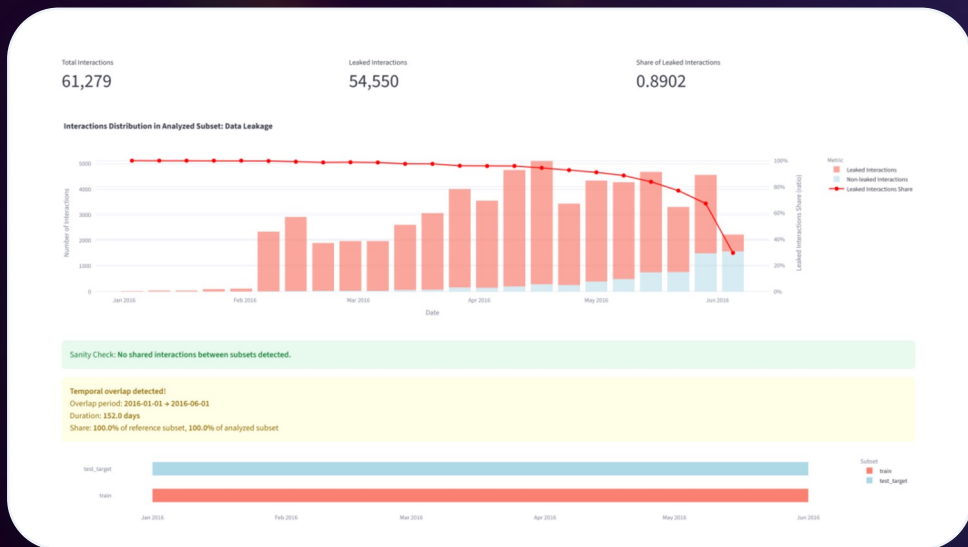


Repeated interactions types



Why care? Consecutive repeats may signal logging issues or may need aggregation to improve RS performance.

Temporal leakage: the future affects evaluation



Data leakage in Diginetica under leave-one-out split

100%

train-eval period
overlap

89%

of targets leak

SPLITLIGHT CHECKS

- Shared interactions between train & eval
- Temporal overlap of their time ranges
- Share of eval targets affected by leakage

Why matters? training data contains information unavailable at real prediction time inflating offline metrics and biasing comparisons.

Cold-start exposure

Cold users / items never seen in training test a different ability than warm-start recommendation.

Splitting affects the rate

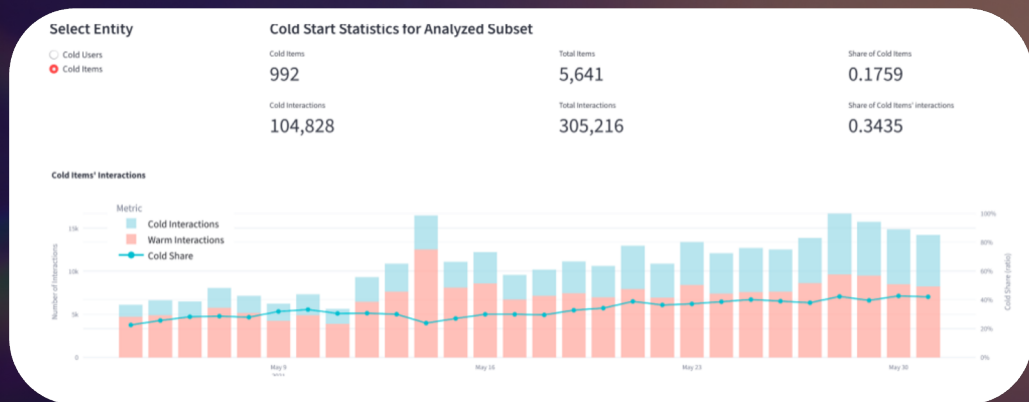
Different strategies induce very different cold-start shares

Uncontrolled = misaligned

Splitting protocol may drift from the real deployment scenario.

Cold start may distort ranking

A lot of cold start data in evaluation could quietly favor some class of models, e.g. embedding-based methods over content / hybrid.

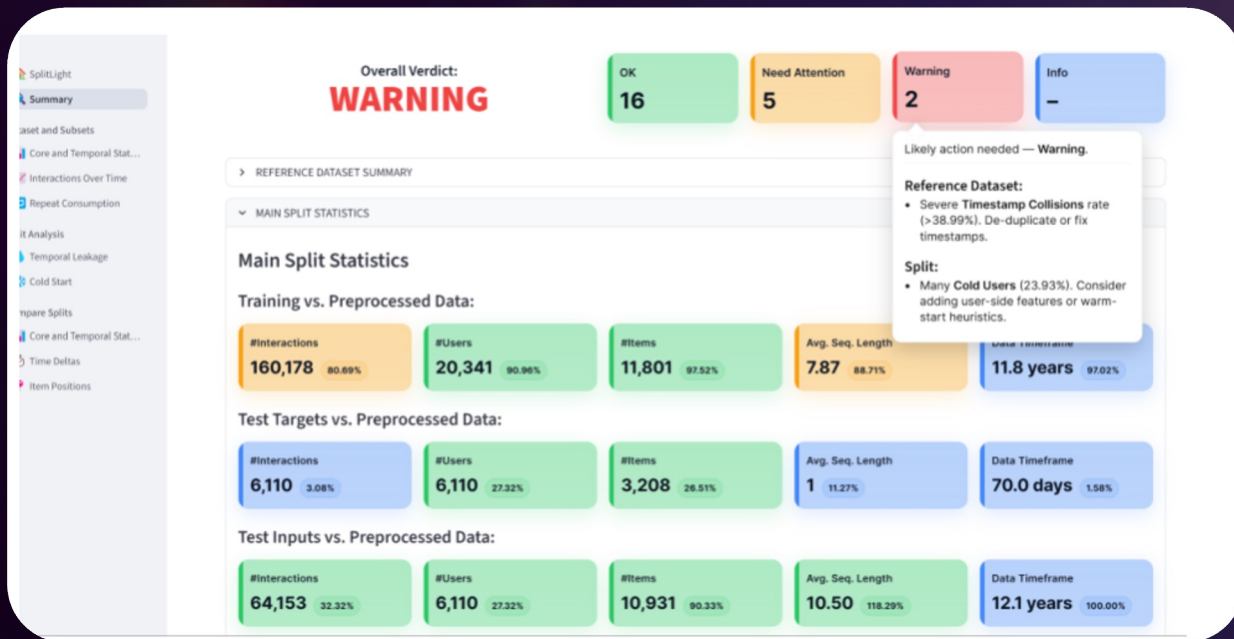


Cold items in the Diginetica test target subset: the share grows over time from ~20% to ~40% (34.35% overall).



What is analysed? Analyse counts and shares of cold users / items and tracks how cold exposure shifts over time.

Summary dashboard: fast health check



See dataset and split statistics on a single page

Get overall data health verdict along with issues list and proposed actions



Entry point: spot high-risk issues, then drill into detailed pages.

Features of widely-used datasets revealed

WHAT THE AUDIT REVEALED

Dataset	#Users	#Items	#Interact.	Avg. Len.	Density (%)	Consec. Repeats (%)	Repeated Interact. (%)
Beauty [18]	22 363	12 101	198 502	8.88	0.07	0.00	0.00
Diginetica [4]	310 324	122 993	1 235 380	3.98	0.003	5.85	13.62
Dressipi [14]	1 000 000	23 496	4 743 820	4.74	0.02	7.25	13.39
ML-1M [7]	6 040	3 706	1 000 209	165.60	4.47	0.00	0.00
ML-20M [7]	138 493	26 744	20 000 263	144.41	0.54	0.00	0.00
Zvuk [23]	20 000	391 322	10 867 482	543.37	0.14	21.50	68.17

Dataset	End Date	Length (years)	Mean Δt	Median Δt	Mean User Lt.	Median User Lt.	Ts. Collis. (%)
Beauty	2014-07-23	12.11	69.65 d	4.00 d	1.50 y	1.00 y	38.9931
Diginetica	2016-06-01	0.42	1.79 m	1.02 m	5.34 m	2.86 m	0.0003
Dressipi	2021-05-31	1.42	10.02 m	45.57 s	37.50 m	1.75 m	0.0139
ML-1M	2003-02-28	2.84	13.85 h	0.00 s	94.99 d	0.05 d	52.8935
ML-20M	2015-03-31	20.22	1.37 d	11.00 s	196.59 d	0.04 d	20.9617
Zvuk	2023-04-16	0.25	1.72 h	0.25 s	38.76 d	32.29 d	0.2195

MovieLens long timeframe is misleading

Median user lifetime ~1 hour — a single session despite long timeframe.

Severe timestamp collisions

ML-1M: >50% of interactions — within-user order is ambiguous.

Zvuk is repeat-heavy

68% repeats; removing consecutive ones raises median Δt 0.25 s $\rightarrow$ 15 s.

Sessions, not preferences

Diginetica & Dressipi suit session-based, not long-term modeling.

Impact on RS evaluation

down to - 9.5%

Shuffling collision order

LOO: re-ordering tied-timestamp events drops NDCG@10 (model memorized an artificial event order)

up to + 60%

Keeping consecutive repeats

GTS $q_{0.9}$: easy-to-memorize repeats inflate the metric and bias models toward short-term signals.

down to - 20%

Including cold items

Cold items in eval cut NDCG@10 by 1.2–20%, scaling with the cold-target share.

Setup: SASRec, NDCG@10, 5 seeds, LOO and GTS Last splits



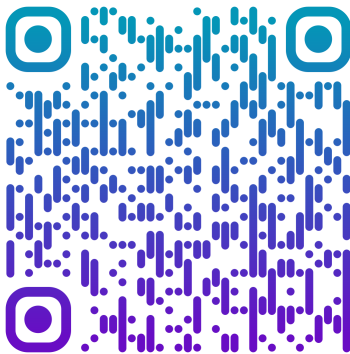
Results under the “same” split are not comparable across papers unless detailed curation protocol is reported.



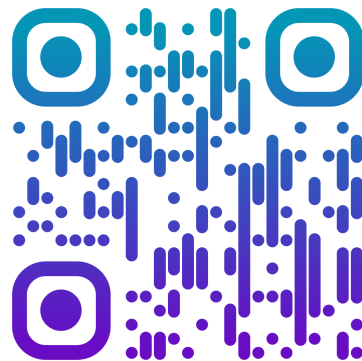
SplitLight: Links



[GitHub](#)



[Demo Video](#)



[arXiv](#)



Thank you for your attention!

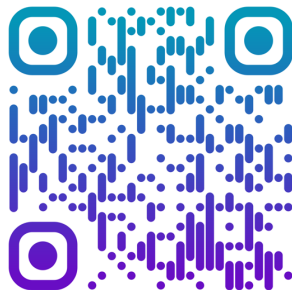
We are open for collaboration!

Email:
alexsl.vasilev@yandex.ru

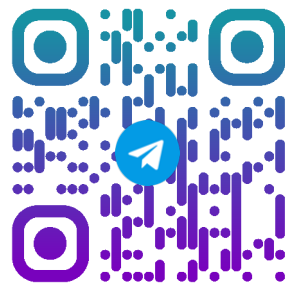
Telegram:
[@AleXXL1986](https://t.me/AleXXL1986)



https://t.me/recsys_themes



GitHub: sb-ai-lab



@sb-ai-lab
Telegram Chanell



For science
collaboration

